

Assignment 3: Used Car Price Prediction - CarDekho

Objective

The primary goal of this project was to develop a robust machine learning model capable of predicting the selling price of used cars based on historical data from the CarDekho dataset.

Data Preparation

To ensure data quality and relevance, the following preprocessing steps were implemented:

- **Filtering:** The dataset was narrowed down to include only "Petrol" vehicles to focus the model's scope.
- **Cleaning:** Duplicate entries were identified and removed to prevent model bias.
- **Feature Engineering:** A new feature, `car_age`, was created by subtracting the vehicle's manufacturing year from the current reference year of 2026.

Exploratory Data Analysis (EDA)

Key insights derived from the initial data exploration include:

- **Price Distribution:** Selling prices exhibit a right-skewed distribution, suggesting the presence of a limited number of high-value luxury vehicles.
- **Transmission Impact:** Vehicles equipped with automatic transmissions generally command significantly higher resale prices compared to their manual counterparts.
- **Depreciation Trends:** Both `car_age` and `km_driven` demonstrate a strong negative correlation with `selling_price`, confirming that older vehicles and those with higher mileage lose value predictably.

Model Performance

The predictive modeling phase involved two distinct iterations:

- **Baseline Model:** A standard Linear Regression model was established as the initial benchmark.
- **Improved Model:** To address the right-skewed nature of the target variable, a log-transformation was applied to `selling_price`. This normalization led to a more consistent distribution and resulted in noticeably improved evaluation metrics across the board.

Conclusion

The final model successfully utilizes critical features—including `car_age`, `km_driven`, fuel type, transmission, and owner history—to provide accurate vehicle valuations. The implementation of target variable transformation proved essential in enhancing the overall predictive accuracy of the system.